Correlation Modelling for Worst-of and Best-of Options: Statistical Tests on the LVMH–Hermès Dependence

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Abstract

Worst-of and best-of options pay on the minimum or the maximum of the performances of a basket, and their price depends on the dependence between the underlyings as much as on their volatilities. This report prepares a *non-stochastic* correlation model for a basket of two French stocks, LVMH and Hermès: the pricing correlation is a deterministic function of the history known at the pricing date, so that the same inputs always give the same price. Before choosing that function, 21 years of daily data (5,588 returns, January 2005 to October 2026) are put through a battery of tests. The correlation does not depend on the return horizon (0.60 from 1 to 60 days), so daily returns can be used. It is not constant: equality across years is rejected ($p \approx 10^{-90}$), the yearly value ranges from 0.10 to 0.85 with a level shift around 2015 (0.47 before, 0.74 after), and the Engle–Sheppard test rejects a constant conditional correlation ($p \approx 10^{-5}$). On GARCH-standardised residuals, a mean-reverting DCC model beats both the constant (likelihood ratio 443) and the integrated EWMA specification (likelihood ratio 60, $p \approx 10^{-14}$); its persistence $a + b = 0.9915$ gives a half-life of 82 trading days, so that over a one-year product only 41% of today's deviation from the long-run correlation survives on average. The joint tails are fatter than under a Gaussian dependence, mostly on the upside, and the correlation is higher in high-volatility regimes. These results set the list of candidate models for the next sections. This paper is linked to my GitHub, which can be found on my portfolio[1], and all scripts are available in the same repository.

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[1] https://portfolio-eliasbenguedda.fr

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1 Introduction

1.1 Motivation

A worst-of (best-of) option on a basket of n stocks pays on the worst (best) of the performances S_T^i/S_0^i. These payoffs are the building blocks of most multi-asset structured products: worst-of autocallables, worst-of reverse convertibles and best-of calls. Each vanilla option on a single underlying identifies the volatility of that underlying, but no listed instrument identifies the correlation between two single stocks with any liquidity. The pricing correlation therefore has to be estimated from history, and the estimator is a modelling choice with a direct price impact.

A pricing desk also needs *reproducibility*: a structured product priced twice with the same inputs must have the same price, for instance when a data error forces a recalculation. This rules out correlation models whose pricing correlation is itself random, such as a correlation following its own diffusion, and motivates a *non-stochastic* model:

$$\rho_{\text{pricing}}(t_0, T) = f\big((r_s)_{s \le t_0},\, T\big), \tag{1}$$

a deterministic function of the returns observed up to the pricing date t_0, possibly depending on the maturity T of the product. The historical correlation over a window, its exponentially weighted version (EWMA) and the forecast of a GARCH-type conditional correlation model are all of this form.

1.2 Scope of this report

Choosing f in (1) requires knowing how the dependence between the underlyings actually behaves. This first part of the study is purely statistical: on LVMH and Hermès, two luxury stocks listed in Paris, it tests whether the correlation depends on the return horizon, whether it is constant, whether it mean-reverts and at which speed, and whether a Gaussian dependence describes the joint extremes. Each test is chosen for its consequence on the model; the models themselves and the pricing of worst-of and best-of options will be the subject of the following sections.

1.3 Organisation of the paper

Section 2 recalls the payoffs and how the correlation enters their price. Section 3 describes the data. Section 4 tests the marginals, Section 5 the dependence of the correlation on the return horizon, Section 6 its constancy, Section 7 compares constant, integrated and mean-reverting conditional correlation models, Section 8 measures the mean reversion without a model and Section 9 studies asymmetry and tails. Section 10 draws the consequences for the model, Sections 11 and 12 describe the implementation and the limitations, and Section 13 concludes.

2 Worst-of and Best-of Options

2.1 Payoffs

Let $X_T^i = S_T^i/S_0^i$ be the performance of stock i and $m_T = \min_i X_T^i$, $M_T = \max_i X_T^i$. The worst-of and best-of calls and puts of strike K pay

$$(m_T - K)^+, \qquad (K - m_T)^+, \qquad (M_T - K)^+, \qquad (K - M_T)^+. \tag{2}$$

For two assets, $m_T + M_T = X_T^1 + X_T^2$ and the identity $(m_T - K)^+ + (M_T - K)^+ = (X_T^1 - K)^+ + (X_T^2 - K)^+$ (and the same for puts) shows that the worst-of and the best-of option of the same type add up to the two vanilla options, which do not depend on the correlation: whatever the correlation adds to one, it takes from the other.

2.2 Role of the correlation

In a multi-asset Black–Scholes model, $\mathrm{d}S^i_t/S^i_t = (r - q_i)\,\mathrm{d}t + \sigma_i\,\mathrm{d}W^i_t$ with $\mathrm{d}\langle W^i, W^j\rangle_t = \rho_{ij}\,\mathrm{d}t$, the dependence between the terminal values is a Gaussian copula with correlation matrix (ρ_{ij}). The spread between the two performances shrinks as the correlation rises: $\mathbb{E}\,|X^1_T - X^2_T|$ decreases with ρ, so the worst performance rises and the best one falls. Worst-of calls are therefore worth more, and worst-of puts less, when the correlation is high; best-of options move the other way. For two assets, the call on the minimum and on the maximum have a closed form [2], which will serve to validate the Monte Carlo pricer in the next part. A worst-of put, the option sold by the investor in a worst-of autocallable, is *short correlation* for the buyer of the note: a correlation estimate that is too high underprices the risk.

The Gaussian copula has two consequences that the tests of this report check: the dependence is summarised by a single number per pair, and the joint extremes are as rare as in a bivariate normal distribution (no tail dependence).

3 Data

3.1 Source and adjustments

Daily closes of LVMH (`MC.PA`) and Hermès (`RMS.PA`) on Euronext Paris are downloaded from Yahoo Finance with the `yfinance` library [1] by the script `Data/Equity/download_data.py`, from 3 January 2005 to 30 October 2026 (5,589 trading days, no missing value). Yahoo data is free but unofficial and its history can be revised, so the files are frozen in the repository and every number of this report is reproducible.

Two series are stored per stock. The *adjusted close*, corrected for dividends and splits, is used for every statistic on returns, so that an ex-dividend date is not read as a price drop. The *raw close* is the underlying of the payoffs (price return); dividends will enter the pricing through the forwards. The log return is $r^i_t = \ln(S^i_t/S^i_{t-1})$ on the adjusted close.

3.2 Synchronicity

Both stocks close at the same time on the same exchange, so their daily returns are synchronous. For pairs listed on different time zones, a European close and an American close are separated by several hours of news, and the correlation of daily returns is biased downwards; multi-day returns are then used. Section 5 checks that this is not needed here.

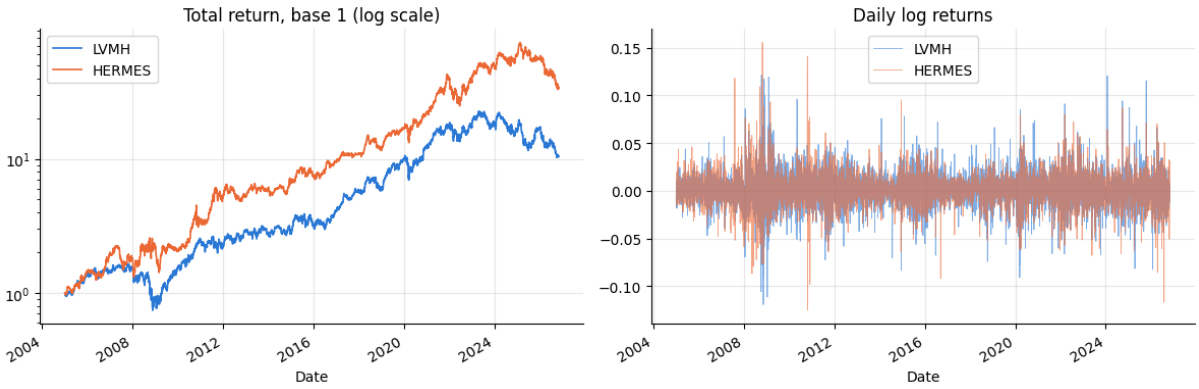


Figure 1: Total return of LVMH and Hermès since January 2005 (log scale) and daily log returns.

	Vol (ann.)	Skew.	Exc. kurt.	JB p	LB r p	LB r^2 p	ARCH-LM p
LVMH	28.3%	0.15	4.9	$< 10^{-300}$	0.018	10^{-154}	6×10^{-63}
Hermès	27.3%	0.15	6.3	$< 10^{-300}$	0.23	4×10^{-171}	7×10^{-89}

Table 1: Marginal statistics of the daily log returns, 2005–2026. JB: Jarque–Bera normality test; LB: Ljung–Box test with 10 lags on returns and squared returns; ARCH-LM: Engle's test with 5 lags.

4 Marginals

4.1 Tests

Three properties of each return series are tested (Table 1): normality with the Jarque–Bera test [3], absence of autocorrelation of the returns and of the squared returns with the Ljung–Box test [4], and absence of conditional heteroskedasticity with Engle's ARCH-LM test [5].

4.2 Results

Both stocks have an annualised volatility of about 28%, a small positive skewness and an excess kurtosis of 5 to 6: normality is rejected without ambiguity. The returns themselves are nearly unpredictable (Ljung–Box $p = 0.018$ for LVMH, 0.23 for Hermès), while the squared returns are strongly autocorrelated: volatility comes in clusters.

A GARCH(1,1) with Student-t innovations [6, 7] is fitted to each stock (Table 2). The persistence $\alpha + \beta$ is 0.990 for LVMH and 0.996 for Hermès, and the degrees of freedom of about 5 confirm fat-tailed innovations. The standardised residuals $\varepsilon_t^i = (r_t^i - \mu_i)/\sigma_{i,t}$ remove the volatility clusters.

	ω	α	β	$\alpha + \beta$	ν
LVMH	0.037	0.050	0.939	0.990	5.1
Hermès	0.019	0.053	0.943	0.996	5.1

Table 2: GARCH(1,1) with Student-t innovations, returns in percent: $\sigma_t^2 = \omega + \alpha\,(r_{t-1} - \mu)^2 + \beta\,\sigma_{t-1}^2$.

4.3 Consequence for the dependence tests

Volatility clustering matters for the correlation. When volatilities move through time, tests of the constancy of the correlation applied to raw returns over-reject, and the sample correlation mechanically rises in volatile periods [17]. Every dependence test below is therefore run on the raw returns and, where it matters, on the GARCH residuals ε_t. On the full sample the two correlations are close: 0.605 on raw returns and 0.609 on residuals.

5 Correlation and Return Horizon

The correlation of non-overlapping h-day log returns is computed for h from 1 to 60 days, with the 95% interval given by the Fisher transform: $\operatorname{artanh} \hat{\rho}$ is approximately normal with variance $1/(n-3)$ [9]. A correlation that increases with the horizon would signal non-synchronous trading or a lead–lag effect (the Epps effect, [10]).

Horizon (days)	n	$\hat{\rho}$	95% low	95% high
1	5,588	0.605	0.588	0.621
2	2,794	0.619	0.595	0.641
5	1,117	0.599	0.560	0.636
10	558	0.609	0.554	0.658
20	279	0.556	0.470	0.632
60	93	0.605	0.458	0.720

Table 3: Correlation of non-overlapping h-day log returns.

The correlation is flat in the horizon, about 0.60 from 1 to 60 days, and every interval contains the daily value (Table 3). There is no synchronicity bias: *daily returns are used*, which gives five times more observations than weekly returns and a correspondingly more precise estimator.

6 Is the Correlation Constant?

6.1 Equality across years

Split the sample into K calendar years with n_k observations and sample correlation $\hat{\rho}_k$. Under independent normal observations, $z_k = \operatorname{artanh} \hat{\rho}_k$ is approximately $\mathcal{N}(\operatorname{artanh} \rho_k, 1/(n_k - 3))$ and, under $H_0 : \rho_1 = \cdots = \rho_K$,

$$Q = \sum_{k=1}^{K} (n_k - 3)(z_k - \bar{z})^2 \sim \chi^2_{K-1}, \qquad \bar{z} = \frac{\sum_k (n_k - 3)\, z_k}{\sum_k (n_k - 3)}. \tag{3}$$

With $K = 22$ years, $Q = 490$ on raw returns ($p = 2 \times 10^{-90}$) and $Q = 429$ on GARCH residuals ($p = 10^{-77}$): removing the volatility clusters barely changes the conclusion. The yearly correlations (Table 4) range from 0.10 in 2014 to 0.85 in 2022. The year 2014 is that of the settlement of LVMH's stake in Hermès, which may explain an idiosyncratic decoupling; the low 2017 value (0.38) is not explained here.

Year	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015
$\hat{\rho}$ returns	0.41	0.46	0.57	0.52	0.55	0.42	0.51	0.43	0.54	0.10	0.76
$\hat{\rho}$ residuals	0.41	0.39	0.55	0.57	0.53	0.49	0.50	0.44	0.52	0.23	0.74
Year	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026
$\hat{\rho}$ returns	0.73	0.38	0.72	0.75	0.77	0.77	0.85	0.82	0.83	0.65	0.67
$\hat{\rho}$ residuals	0.70	0.39	0.69	0.73	0.78	0.77	0.85	0.81	0.84	0.67	0.68

Table 4: Yearly correlations of daily returns and of GARCH residuals (2026: January to October).

Figure 2 shows a level shift around 2015: the correlation is 0.47 over 2005–2014 and 0.74 over 2015–2026, with fluctuations of ± 0.2 around each level at the quarterly scale. The full-sample value of 0.61 is representative of neither period. For a pricing correlation, the estimation window or the weighting of the observations is therefore a first-order choice: a historical correlation over the whole sample would mix two regimes. Over the last year the correlation is 0.67, over the last three years 0.70.

6.2 Constancy of the conditional correlation

The yearly test compares unconditional correlations. The test of Engle and Sheppard [11] addresses the conditional correlation directly. The standardised residuals are whitened with the

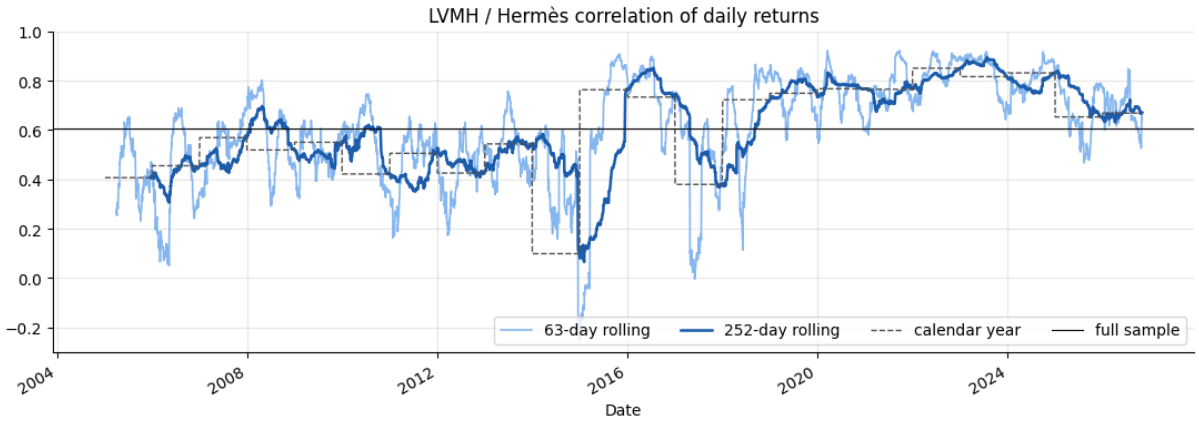


Figure 2: Rolling correlation of daily returns over 63 and 252 days, yearly values and full-sample value.

constant correlation matrix, $u_t = L^{-1}\varepsilon_t$ with $\bar{R} = LL^\top$ the Cholesky factorisation of the sample correlation of ε_t. Under a constant conditional correlation, u_t^1 and u_t^2 are uncorrelated at every date and the cross product $Y_t = u_t^1 u_t^2$ has mean zero and no predictability. The regression

$$Y_t = \delta_0 + \delta_1 Y_{t-1} + \cdots + \delta_s Y_{t-s} + \eta_t \tag{4}$$

gives the statistic $\hat{\delta}^\top X^\top X \hat{\delta} / \hat{\sigma}_\eta^2 \sim \chi^2_{s+1}$ under the null.

Lags s	Statistic	df	p-value
1	23.3	2	9×10^{-6}
5	33.1	6	1×10^{-5}
10	47.1	11	2×10^{-6}

Table 5: Engle–Sheppard test of constant conditional correlation on the GARCH residuals.

The constant conditional correlation is rejected at every lag (Table 5): past co-movements predict future co-movements, which is what a dynamic correlation model exploits.

7 Constant, Integrated or Mean-Reverting Correlation

7.1 Three specifications

Following the two-step approach of Engle [12], the conditional correlation is modelled on the GARCH residuals through a pseudo-correlation matrix Q_t, with $\rho_t = Q_{12,t}/\sqrt{Q_{11,t}\,Q_{22,t}}$. Three nested specifications are compared:

$$\text{CCC [8]:} \quad Q_t = \bar{Q}, \tag{5}$$

$$\text{EWMA [13]:} \quad Q_t = (1-\lambda)\,\varepsilon_{t-1}\varepsilon_{t-1}^\top + \lambda\, Q_{t-1}, \tag{6}$$

$$\text{DCC [12]:} \quad Q_t = (1-a-b)\,\bar{Q} + a\,\varepsilon_{t-1}\varepsilon_{t-1}^\top + b\, Q_{t-1}, \tag{7}$$

with $\bar{Q}$ the sample covariance of ε_t. The EWMA is the DCC on the boundary $a+b=1$, without a long-run target, and the CCC is the DCC with $a=0$. All three are deterministic functions of the past and are therefore candidates for (1). The parameters maximise the correlation part of the Gaussian log-likelihood,

$$\ell_C = -\frac{1}{2}\sum_t \left[\ln(1-\rho_t^2) + \frac{(\varepsilon_t^1)^2 + (\varepsilon_t^2)^2 - 2\rho_t\,\varepsilon_t^1\varepsilon_t^2}{1-\rho_t^2} - (\varepsilon_t^1)^2 - (\varepsilon_t^2)^2\right]. \tag{8}$$

7.2 Forecasts over the life of a product

The three models differ most in their forecasts. The CCC forecast is constant. The EWMA is integrated: its forecast of Q_{t+h} is Q_{t+1} for every h, so the correlation of today is used for the whole life of the product. The DCC reverts to the long-run correlation $\bar{\rho}$; with the approximation of Engle and Sheppard [11],

$$\mathbb{E}_t[\rho_{t+h}] \approx \bar{\rho} + (a+b)^{h-1}(\rho_{t+1} - \bar{\rho}), \tag{9}$$

and the average correlation over a product of H days is

$$\frac{1}{H}\sum_{h=1}^{H} \mathbb{E}_t[\rho_{t+h}] \approx \bar{\rho} + w(H)\,(\rho_{t+1} - \bar{\rho}), \qquad w(H) = \frac{1-(a+b)^H}{H\,(1-a-b)}. \tag{10}$$

7.3 Results

Model	ℓ_C	Parameters	AIC	BIC	Estimates
CCC	1293.2	1	-2584	-2578	$\bar{\rho} = 0.609$
EWMA	1484.8	1	-2968	-2961	$\lambda = 0.977$
DCC	1514.7	3	-3023	-3004	$a = 0.031$, $b = 0.961$, $a+b = 0.9915$

Table 6: Conditional correlation models fitted on the GARCH residuals (5,588 days). The DCC count includes the targeted $\bar{\rho}$.

H_0	H_1	LR	df	p-value
CCC (constant)	EWMA	383.3	1	2×10^{-85}
CCC (constant)	DCC	443.1	2	6×10^{-97}
EWMA (integrated, $a+b=1$)	DCC	59.8	1	1×10^{-14}

Table 7: Likelihood ratio tests between the conditional correlation models.

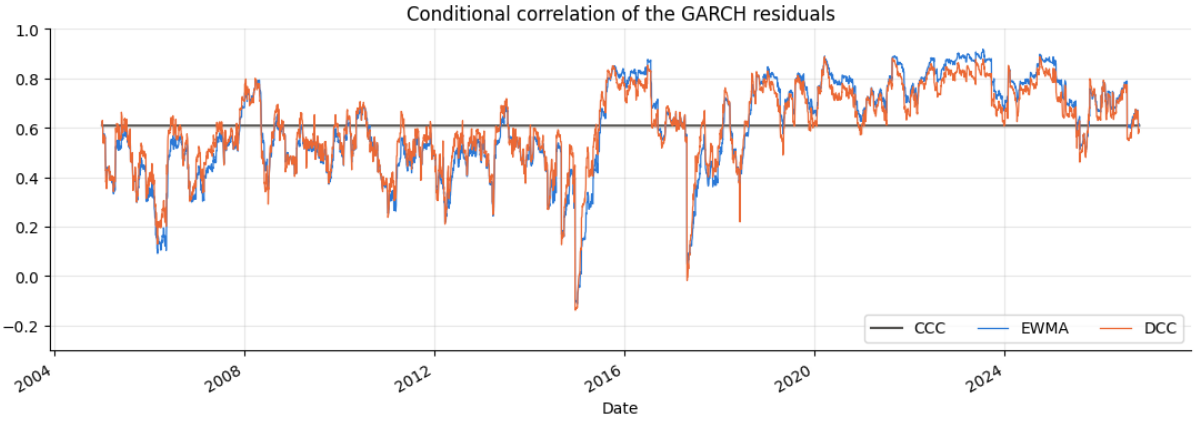


Figure 3: Conditional correlation of the GARCH residuals under the CCC, EWMA ($\lambda = 0.977$) and DCC models.

Both dynamic models beat the constant correlation by a wide margin, and the DCC beats the EWMA (Tables 6 and 7). The paths are close (Figure 3): the two models differ less in their filtered correlation than in their forecast. The conditional correlation is *persistent but*

mean-reverting: $a + b = 0.9915$ gives a half-life of $\ln 2 / |\ln(a + b)| = 82$ trading days, about four months.

Formula (10) turns this into weights on today's deviation from the long-run level:

Product maturity	3 months	1 year	3 years
Weight $w(H)$ of the current deviation	0.78	0.41	0.16

Over a one-year product, only 41% of the current deviation survives on average; over three years, 16%. An EWMA, whose forecast is flat, keeps 100% at every maturity. The fitted EWMA parameter $\lambda = 0.977$ (an observation half-life of 30 days) is also well above the 0.94 of RiskMetrics [13], which was calibrated for one-day risk forecasts, not for a correlation that must hold over the life of a product.

8 Mean Reversion of the Realised Correlation

The DCC result depends on the model. A model-free check uses the rolling correlation $\hat{\rho}_t^{(w)}$ over a window of w days. On its Fisher transform, the augmented Dickey–Fuller test [14] tests the null of a unit root (no mean reversion), and the first-order autocorrelation between consecutive non-overlapping windows measures the persistence from one window to the next.

Window (days)	Mean	Std	Min	Max	ADF p	AR(1) non-overlapping
63	0.61	0.21	−0.25	0.92	0.0004	0.59
126	0.61	0.19	−0.08	0.90	0.054	0.60
252	0.61	0.17	0.07	0.89	0.13	0.70

Table 8: Rolling correlation of daily returns: level, dispersion and mean reversion.

At the quarterly window, the unit root is rejected ($p = 0.0004$) and two consecutive quarters share an autocorrelation of 0.59 only: short-term correlation reverts (Table 8). At the yearly window the test loses power (overlapping windows, about 20 independent years) and the persistence is higher (0.70). The data show two time scales: a slowly moving long-run level, which moved in 2015, and fast fluctuations around it. A pricing correlation has to answer two questions: which long-run level, and how fast the current value reverts to it.

9 Asymmetry and Tails

The multi-asset Black–Scholes model imposes a Gaussian copula. Three statistics test it, each compared to its value under a bivariate normal distribution with the same correlation, since a Gaussian has non-trivial values of these statistics too.

9.1 Exceedance correlations

Following Longin and Solnik [15], the exceedance correlation at threshold q is the correlation of the standardised returns conditional on both being below $-q$ standard deviations (down) or above q (up). For a Gaussian it decreases with q and is symmetric; the benchmark is computed by simulation (2×10^6 draws).

The exceedance correlations are above the Gaussian benchmark at every threshold and on both sides (Table 9, Figure 4). Part of the gap comes from the time variation of the correlation itself: a mixture of Gaussians with different correlations has fatter joint tails than a single Gaussian. On the GARCH residuals the up side stays high at 1.5 standard deviations (0.47 against 0.22) while the down side comes back to the Gaussian level (0.24 against 0.21). For this

		Raw returns			GARCH residuals		
q (sd)	Side	n	Data	Gaussian	n	Data	Gaussian
0	down	1,996	0.51	0.36	1,994	0.50	0.37
	up	2,029	0.51	0.36	2,029	0.44	0.36
0.5	down	862	0.46	0.30	917	0.45	0.31
	up	819	0.52	0.30	898	0.42	0.31
1	down	349	0.33	0.25	364	0.37	0.25
	up	288	0.45	0.25	325	0.42	0.26
1.5	down	135	0.28	0.20	143	0.24	0.21
	up	119	0.36	0.22	109	0.47	0.22

Table 9: Exceedance correlations against a Gaussian dependence with the same correlation.

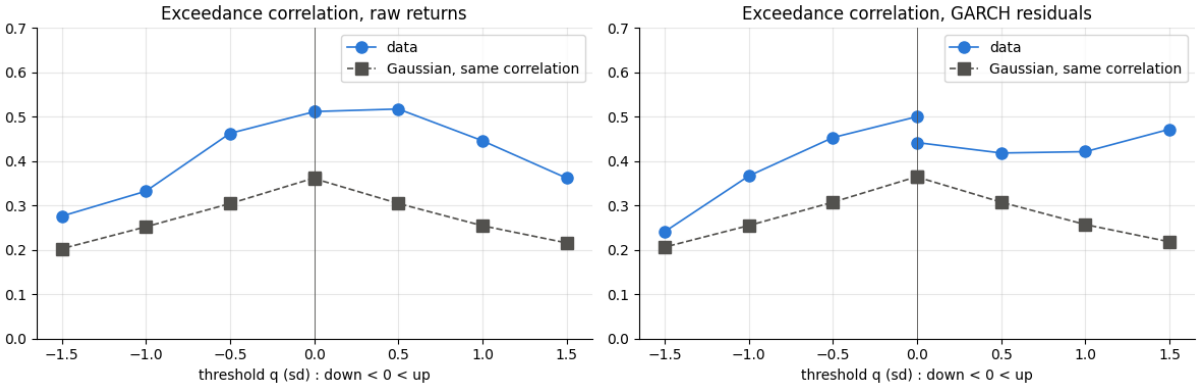


Figure 4: Exceedance correlations (left of zero: both returns below $-q$; right: both above q).

pair there is no sign of a correlation that rises in sell-offs; the strongest co-movements are joint rallies, consistent with sector-wide news for luxury stocks.

9.2 Tail dependence

On the ranks U_t^i of the returns, the empirical lower tail dependence at level q is $\hat{\lambda}_L(q) = \#\{t : U_t^1 < q, U_t^2 < q\}/(nq)$, and $\hat{\lambda}_U(q)$ is defined in the same way on the upper tail [16]. For a Gaussian copula $\lambda_L(q) \to 0$ as $q \to 0$.

	Raw returns		GARCH residuals		
q	Lower	Upper	Lower	Upper	Gaussian
10%	0.51	0.46	0.50	0.41	0.39
5%	0.41	0.39	0.43	0.36	0.31
1%	0.20	0.43	0.30	0.27	0.19

Table 10: Empirical tail dependence against a Gaussian copula with correlation 0.605. At 1% the lower tail of the raw returns rests on 11 joint observations and the upper tail on 24.

Both tails are above the Gaussian at 10% and 5% (Table 10). At 1% the estimates rest on a few dozen joint observations at most and are not conclusive on their own; on raw returns the upper tail stands out (0.43 against 0.19), in line with the exceedance correlations.

9.3 Volatility and market regimes

The 21-day rolling correlation is averaged over the days in the top and bottom 20% of the average individual volatility, and over the days when the equally weighted basket fell or rose over the previous 21 days. Conditioning on high volatility raises the sample correlation even when the true correlation is constant [17]; the benchmark applies the same statistics to simulated returns with a constant correlation and the realised volatility path of the data (average over 50 simulations).

Regime	Data	Constant-correlation benchmark
High volatility (top 20%)	0.66	0.62
Low volatility (bottom 20%)	0.45	0.55
Down market (21-day basket < 0)	0.64	0.59
Up market (21-day basket > 0)	0.58	0.59
All days	0.61	0.59

Table 11: Mean 21-day rolling correlation by regime.

The correlation is 0.21 higher in high-volatility regimes than in low-volatility ones, against 0.08 for the benchmark (Table 11): about 40% of the gap is the conditioning bias, the rest is a genuine link between correlation and volatility. Down markets show a slightly higher correlation than up markets (0.64 against 0.58, with a benchmark of 0.59 on both sides). The correlation is thus linked to the volatility level, which matters for worst-of products since their value is concentrated in the scenarios where the volatility is high.

10 Synthesis and Consequences for the Model

Test	Result	Consequence for the model
Marginals	Fat tails, strong volatility clustering	Test the dependence on GARCH residuals; the volatility input matters as much as the correlation
Horizon	Correlation flat from 1 to 60 days	Daily returns, no synchronicity correction
Constancy	Rejected (Fisher, Engle–Sheppard); shift around 2015	The window or the weighting is first-order; a 20-year average is biased
CCC / EWMA / DCC	DCC preferred; $a+b = 0.9915$, half-life 82 days; EWMA $\lambda = 0.977$	A mean-reverting forecast over the product horizon is supported against a flat EWMA forecast
Mean reversion	Short-term correlation reverts; slow long-run level	Two scales: long-run level and reversion speed
Tails and regimes	Joint tails fatter than Gaussian, mostly upside; higher correlation in high volatility	A Gaussian copula understates joint extremes; correlation linked to the volatility level

Table 12: Summary of the tests.

The tests (Table 12) set the list of non-stochastic candidates for (1), which the next part of the study will build and compare:

- *Historical correlation* over a window (full sample, one year, three years, since the 2015 shift, or equal to the maturity): the benchmark, whose dependence on the window measures the estimation risk;

- *EWMA*, on raw returns or on GARCH residuals, with λ chosen by likelihood or by the forecast error over the product horizon;
- *DCC forecast*: the average correlation over the life of the product, formula (10), which is the specification preferred by the likelihood and makes the pricing correlation depend on the maturity;
- *blend of a short-term and a long-term estimate* with maturity-dependent weights, a simple version of the DCC forecast;
- *regime-switching forecast* [18]: the filtered regime probabilities give a deterministic expected correlation, motivated by the 2015 shift and by the link with volatility;
- *Student-t copula* instead of the Gaussian one: a fixed number of degrees of freedom adds tail dependence, motivated by Section 9.

The candidates will be ranked out of sample on two criteria: the error of the forecast against the correlation realised over the life of the product, and the impact on the prices of worst-of and best-of options.

11 Implementation

- `Data/Equity/download_data.py`: downloads the raw and adjusted closes and the dividends of LVMH and Hermès from Yahoo Finance into `Data/Equity/`;
- `Models/correlation_tests.py`: all the tests of this report (marginal tests, horizon, Fisher equality test, Engle–Sheppard, CCC/EWMA/DCC fits and likelihood ratios, mean reversion, exceedance correlations, tail dependence, regimes with their Gaussian or constant-correlation benchmarks). The recursions on Q_t are linear filters, so a likelihood evaluation over 5,588 days takes about a millisecond;
- `Models/correlation.ipynb`: the notebook of this report;
- `Reports/Correlation/make_figures.py`: extracts the figures from the executed notebook.

```
import pandas as pd
import correlation_tests as ct

raw = pd.read_csv("Data/Equity/LVMH_HERMES_Daily.csv", index_col="Date",
    parse_dates=True)
px = raw[["LVMH_AdjClose", "HERMES_AdjClose"]].set_axis(["LVMH", "HERMES"], axis
    =1)
r = ct.log_returns(px)
eps, garch = ct.garch_standardise(r)                    # GARCH(1,1)-t residuals
models, lr_tests, cond_rho, params = ct.compare_conditional_models(eps)
print(ct.engle_sheppard_test(eps), lr_tests, params["DCC"])
```

12 Limitations

- *Two stocks, one period.* The conclusions hold for LVMH and Hermès over 2005–2026; the asymmetry of the tails in particular (joint rallies rather than joint crashes) is specific to this pair and may not hold for a basket of indices or of stocks from different sectors.

- *Assumptions of the tests.* The Fisher equality test assumes independent normal observations; it is repeated on GARCH residuals, which removes the volatility clusters but not the fat tails. The likelihood ratio of CCC against DCC has a parameter that is not identified under the null and the EWMA lies on the boundary of the DCC parameter space, so the χ^2 p-values are indicative; with ratios of 60 to 443 no correction changes the conclusions.
- *Two-step estimation.* The correlation parameters are estimated given the GARCH fits, with a Gaussian quasi-likelihood although the residuals are Student; the standard errors of the second step are not corrected for the first.
- *In-sample.* All tests use the full sample. The model choice will be validated out of sample.
- *Data.* Yahoo Finance prices are free but unofficial; the adjusted close relies on Yahoo's dividend adjustment.

13 Conclusion

Over 21 years of daily data, the correlation between LVMH and Hermès averages 0.61 but is neither constant nor a random walk. It shifted from about 0.47 to 0.74 around 2015, fluctuates by ± 0.2 at the quarterly scale and reverts with a half-life of about four months. A mean-reverting DCC model is preferred to both a constant correlation and an integrated EWMA, and the joint extremes are more frequent than a Gaussian dependence implies, mostly on the upside. For a non-stochastic pricing correlation, the main question is therefore less how to estimate the correlation of today than how much of it to keep over the life of the product: with the fitted DCC, 41% of today's deviation over one year and 16% over three years.

A Notation

Symbol	Meaning
S^i_t, $X^i_T = S^i_T/S^i_0$	Price and performance of stock i
m_T, M_T	Worst and best performance
r^i_t	Daily log return (adjusted close)
$\sigma_{i,t}$, ε^i_t	GARCH conditional volatility and standardised residual
ρ, ρ_t, $\bar{\rho}$	Correlation, conditional correlation, long-run correlation
Q_t, $\bar{Q}$	Pseudo-correlation matrix of the DCC and its target
λ	EWMA decay
a, b	DCC innovation and persistence parameters
$w(H)$	Weight of the current deviation in the average forecast over H days
$\lambda_L(q)$, $\lambda_U(q)$	Empirical lower and upper tail dependence at level q

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